

SELF-EVOLUTION INTELLIGENCE FRAMEWORK

A Comprehensive Guide to Architectural Research, Model Merging, and Circuit-Wise Evolution

Part I: Universal Rules of Logic & Architecture

Recent research (2025–2026) has identified several "Physical Constants" for Large Language Models. For an evolutionary system like **Project IRONCLAD**, these rules dictate the fitness boundaries of mutated architectures.

1. The Mid-Block Stability Rule

Reasoning is not uniform. The **20%–80% depth zone** (the "Logic Mid-block") is the primary locus for mathematical reasoning and deductive logic. While the first 20% (Embeddings/Facts) and final 20% (Formatting/Instructions) exhibit high plasticity, the Mid-block must remain stable to prevent logic collapse.

2. The MoE Activation Sweet Spot

Mixture-of-Experts (MoE) models consistently beat dense models of equal compute when the **activation rate is tuned to ~20%**. For **Breeding Vat** hybridization, target 4 experts with 2 active per token (4/2) for the highest stability-to-reasoning ratio.

3. Minimum Depth Threshold

For multi-hop reasoning, **depth overrides width**. A model requires a minimum critical depth to resolve latent reasoning chains. If a "shrunk" or pruned model fails on logic, scaling depth by just 2–4 layers is often more effective than adding thousands of neurons to existing layers.

Part II: The Merge Fusion Handbook

Heuristics for merging within the mergefusion workflow. These are proven constants derived from high-dimensional loss landscape analysis.

Method	Subtype/Detail	The "Proven Rule"
Task Arithmetic	Logic Preservation	Scale the Delta vector by $\omega \approx 0.7$. Factors >1.0 trigger "Activation Explosions."
SLERP	Divergent Pairs	Optimal interpolation $t \approx 0.5 \pm 0.1$. Preserves vector magnitude unlike Linear.

Method	Subtype/Detail	The "Proven Rule"
TIES	3+ Models	Trim ratio k must decrease as model size increases (7B: 0.8 70B: 0.3).
DARE	High Entropy	Use DARE-TIES if Sign Agreement is < 60% . Best for generalist "super-merges."
Passthrough	Layer Stacking	Only stack in the 0-20% or 80-100% zones. Never skip/stack the mid-block.
FrankenMoE	Top-P Routing	Superior for Coding/Math. Allows variable active experts based on token difficulty.

Part III: Project-Specific Strategies

Breeding Vat: Hybridization Logic

To hybridize models effectively, use **Interleaved Stacking**. Instead of raw block-copying, calculate the weight average between adjacent layers to create "synthetic bridge layers." This reduces the Perplexity Spike by up to 40%.

Project IRONCLAD: Swarm Dynamics

For programming swarms, prioritize **CWP (Circuit-Wise Pruning)**. Research suggests that "Coding Circuits" are often localized in specific Attention Heads. Pruning the "Chat/Safety" circuits while locking the "Logic/Syntax" circuits produces a hyper-efficient specialized swarm.

Creative Insight: Preserving Traits via CWP
CWP is traditionally used for safety/ethics. By mapping **Formatting Circuits** (e.g., JSON output, specific tone), you can treat these as "Immune Zones" during the evolutionary mutation phase, ensuring that structural outputs remain valid even as logic is "bred" upward.

Part IV: System Gap Analysis & Tool Recommendations

Reviewing the LLM-Self-Evolution-System on GitHub, the following components are recommended for a fully "aware" autonomous pipeline:

Missing: Mechanistic Interpretability Stack

- ◆ **SAE Lens:** Critical for "peeking" into the mental state. It maps activations to Sparse Autoencoder (SAE) latents, allowing you to see if your evolved model is activating the correct "Reasoning Features."
- ◆ **ACDC (Automated Circuit Discovery):** Automates the discovery of specific logic circuits. This replaces manual guessing for CWP "Immune Zones."
- ◆ **TransformerLens:** Enables **Activation Patching**. If Model A is better at math than Model B, you can patch Model A's mid-block into Model B to prove where the intelligence resides.

Missing: Evolutionary Optimization Frameworks

- ◆ **Optuna / Ray Tune:** To move beyond manual weights, you need a hyperparameter framework to manage the evolutionary "trials." It uses Bayesian Optimization to find the TIES/DARE constants that maximize your specific logic-gate benchmarks.
- ◆ **Neuronpedia API:** Allows the system to programmatically compare its internal feature activations against a database of "Gold Standard" features (like Qwen's known reasoning latents).